

Room 12 follow-up: the continuity-tracking question, resolved

2026-09-27

The question

Reviewer AW raised a serious concern (see `room12_summary.tex`): `ratio_amp` computes $s = \sqrt{c^N}$ via an independent principal-branch call at each sampled point, with no continuity tracking across nearby N . If the physically meaningful branch should vary *continuously* with (N, a) (tracking the actual saddle asymptotic), then the observed erratic/reciprocal-bimodal behaviour in Room 7's even- a data might be substantially a branch-cut discontinuity artifact of the tool, not a real feature of A_+/A_- .

The check

Direct computation (Python/mpmath, 150 digits – the genuinely huge dynamic range here, $|c^N| \sim 10^{-96}$ at these (N, a) , needs this much precision; earlier attempts at 50–60 digits returned false “degenerate” flags from an over-strict, non-relative threshold, not a real computational failure), $q = 29$, $N = 277$ to 303 even- a (12 points), comparing:

- **Principal branch:** $s = \sqrt{c^N}$, mpmath's own principal square root, independently at each N (what `ratio_amp` does).
- **Continuity-tracked branch:** at each N , pick whichever of $\pm\sqrt{c^N}$ is closer to the previous point's chosen value.

Result: the continuity-tracked sequence **STILL** oscillates between the same two clusters (≈ 0.013 – 0.015 and ≈ 65 – 78) at essentially every step, exactly as the naive principal-branch sequence does. **The**

reciprocal-bimodal/erratic phenomenon is NOT a branch-tracking artifact; it is a genuine feature of A_+/A_- as N varies at fixed seed $a/29$.

Correction (post-review, Reviewer AX): the specific “2 flip points” claim below is NOT reproducible and is withdrawn. My original run found apparent branch disagreements at $N = 291, 299$; Reviewer AX’s independent reimplementaion of the identical continuity-tracking method found disagreements instead at $N = 283, 287, 289, 301, 303$ (5 points, not 2), with no disagreement at all at $N = 291$ or 299 . Investigating why: comparing the raw magnitude of a rapidly-decaying complex quantity across non-adjacent sample points is a genuinely ambiguous notion of “continuity” here – the underlying $\arg(A_+/A_-)$ sequence decreases only slowly and non-monotonically across consecutive even- a points, so different reasonable implementations of the nearest-branch heuristic (differing in comparison order, precision, or tie-breaking) can disagree on exactly which points flip. **No specific flip-point claim, or the associated “real bug in ratio_amp” framing, is retained.** What IS retained and independently confirmed twice (my own run and Reviewer AX’s independent reimplementaion, cross-checked against the actual compiled `ratio_amp` binary): the oscillation between the two clusters is genuinely present under continuity tracking by any of the tested conventions, so it is not an artifact of the naive per-point principal branch specifically. Whether a canonical, unambiguous continuity convention exists that would resolve individual point assignments cleanly is left open; it does not affect the qualitative conclusion below.

Verdict

Room 7’s and Room 9’s core findings STAND, un-retracted – independently confirmed twice (this note’s own computation, and Reviewer AX’s separate reimplementaion cross-checked against the compiled binary). The concern that motivated this check – that the whole even- a erratic/bimodal phenomenon might dissolve under a mathematically correct continuous branch choice – is answered: **it does not dissolve.** The phenomenon is real under every continuity convention tested. The secondary claim of two specific misclassified points is withdrawn (see correction above); whether `ratio_amp` has any real per-point branch bug, and if so which points it affects, is left genuinely open – a future fix would need a precisely specified, unambiguous continuity convention, not the ad hoc nearest-branch heuristic used here.

Weakest points. Checked only $q = 29$, only 12 points, only even- a . The two flip points found here are a small sample; a systematic re-scan of Room 7's full point set with continuity tracking (not done here, would need porting this logic into `ratio_amp` itself in Rust for the larger N Room 7 actually used) would be the natural next step if this thread is revisited, but is not needed to answer the yes/no question this note set out to resolve.